TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-temperature ceramic marking of metal parts for track-and-trace applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Service		Ceramics	
	Medium Category	Small Category	Medium Category	Small Category
	R&D/Engineering service	Product quality control technology	Materials for machines and structures	heat-resistant materials
Concept of Proposed Technology	Due to the high-temperature processes involved in metal production, it is difficult to establish traceability throughout the entire production chain, which is a key requirement for implementing a smart factory. This is a particular challenge for large parts in heavy industry (high-temperature processes, harsh conditions). This project aims to address this issue by developing a high-temperature ceramic marking that can be applied to metal parts using a (mobile) inkjet-printing system. The main functions of the technology are (i) to create durable, high-contrast and machine-readable markings that withstand temperatures up to 1,300 °C and harsh chemical environments, and (ii) to enable reliable identification and tracking of individual parts at each production step, even after coating of the metal parts. Once this technology has been integrated into production, single-part traceability will be possible, paving the way towards a smart factory and a closed-loop economy. Potential uses include traceability of large and high-value metal components in sectors such as steel production, shipbuilding, heavy machinery, and automotive manufacturing, especially where current marking methods fail due to extreme process conditions.			

Rationale for Joint R&D Project	The technology for single-part traceability of metal parts in harsh production environments is strongly related to German core competencies in high-performance ceramics, functional inks, and industrial printing technology. Since heavy industries such as shipbuilding, steel production, and large machinery manufacturing are key sectors in Korea, there is a strong need to enable smart production and end-to-end traceability throughout the whole process chain. Joint R&D is required because the successful development and commercialization of the technology depend on the combination of both partners' strengths: the research institute contributes advanced know-how in ceramic ink formulation, high-temperature materials, and printing/scanning concepts, while Korean SMEs provide access to relevant industrial use cases, production lines for pilot implementation, and the market knowledge necessary for adapting and scaling the solution to the Korean and wider Asian markets. Only through close collaboration can the technology be validated under real Korean industrial conditions and transferred into commercially viable products and services.
Key Words	1. High-temperature ceramic ink 2. Direct part marking 3. Track-and-trace/ Part traceability 4. Smart manufacturing (Industry 4.0) 5. AI-based process data analytics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	• New high-temperature ink adapted to specific applications • Scale-up of ink production • Development of a mobile drop-on-demand (DoD) inkjet printer for particle-based (ceramic) inks • Evaluation of the new printing technology in a production environment • Detection of regular, deformed and/or coated markings to ensure traceability • Development of a data acquisition and (AI-based) analysis framework to link part and process parameters and support smart factory implementation			
R&D Target Specifications	• Development of a high-temperature ceramic marking for metal parts (temperatures up to 1,300 °C) • Mobile printing technology based on drop-on-demand inkjet printing systems for ceramic particle-based, high-temperature-stable inks • Reproducible ink production on an industrial scale with consistent quality (batch size > 10 L) • Traceability of parts throughout the whole production and life cycle (> 95 %) • Improved product quality and reduced production waste • Marking remains machine-readable after at least one complete process cycle (including heat treatment, surface treatment, cleaning) by at least 95 % • Reduction of scrap and rework related to misidentified parts by at least 99 %			
R&D Plan	The R&D plan is structured into consecutive phases, combining laboratory formulation and testing, pilot-scale validation in industrial environments, and scale-up of ink production and printing technology. Pre-study (month 0-6) • Definition of specifications • Test prints with existing inks on metal parts • Market research			

	• Methodology: literature review, benchmarking of existing inks, preliminary lab tests • Milestone: definition of detailed technical specifications and selection of reference ink systems Development of new ink recipe for marking metal parts (month 7-19) • Development of ink with suitable ceramic pigments, binders and additives for specific use case and industrial conditions • Test prints on metal parts in the lab and production line • Long-term printing tests • Methodology: iterative laboratory-scale formulation, rheology adjustment, adhesion and durability testing under industry-realistic process conditions • Milestone: formulation of at least one ink candidate that fulfills high-temperature stability and printability requirements Printing technology (month 12-20) • Modify and test industrial DoD printhead with new ink (stationary and mobile printers) • Test and vary printing parameters/conditions • Detection of markings on metal samples with optional coatings (corrosion protection, paint) • Methodology: adaptation and testing of industrial DoD printheads with ceramic particle-based inks, optimization of printing parameters to ensure stable jetting and mark quality. • Milestone: demonstration of stable printing with required resolution and contrast on relevant metal substrates. Ink production (month 15–21) • Set up of production equipment (mixers, grinders, etc.) • Scale-up ink production from lab to industrial scale • Methodology: scale-up of mixing and milling processes, quality control of ink batches (viscosity, particle size distribution, stability). • Milestone: reproducible production of ink batches > 10 L meeting defined quality specifications. Integration of printer and scanner (month 18-26) • Integration and test of printing technology in production line • Integration and test of scanner in production line • Testing of track-and-trace functionality in production environment • Milestone: successful end-to-end track-and-trace demonstration in a relevant industrial environment at a Korean SME Smart factory (month 24-30) • Linking of production and part parameters • Collect data in database for analysis using AI • Data analysis, e.g. improving part quality and reducing waste based on collected data • Methodology: implementation of data acquisition and AI-based analysis tools, correlation of process parameters with part quality indicators. • Milestone: proof-of-concept demonstration of data-driven process optimization (e.g. reduction of scrap rate in pilot line)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• High-temperature ink development[1–4] • Concept and support for production line integration of the printer system • Detection and identification of (coated) markings • Ink technology transfer

Expected R&R of Korean SME Partner	• Ink production / scale-up for mass production • Installation of the marking system in the production lines of end users (e.g. furnace manufacturers, press manufacturers, ceramics or metal producers) • Integration of scanner equipment into the production line of an end user • Data acquisition (aggregation of part numbers and production parameters in a time series), (AI-based) production process / time series analysis • Provision and coordination of end-user use cases (e.g. automotive, steel, shipbuilding), including access to real production environments for pilot testing

5. Expected Outcomes/Impact

Expected Outcomes and Impact	<u>Expected technical outcomes</u> Development and validation of a novel, high-temperature and chemically resistant ink system for direct marking of metallic parts. The ink will withstand process temperatures up to 1,300 °C and harsh process environments (e.g. heat treatment, surface treatment, cleaning) without losing legibility. The project aims to increase the Technology Readiness Level from TRL 3 to TRL 7 by demonstrating stable performance in relevant industrial pilot lines at Korean SMEs. <u>Impact on manufacturing, quality and traceability</u> The new ink will enable robust, part-level traceability throughout the entire production chain, including high-temperature and chemically aggressive process steps where conventional markings fail. By reliably linking each individual part to its specific process and production parameters, manufacturers can: • Reduce misidentification and mix-ups of parts by more than 99 %. • Enable systematic root-cause analysis and continuous process optimization based on real production data. • Improve outgoing quality and reduce scrap, rework, and recall risks. <u>Economic and commercial impact</u> For Korean SMEs, the technology will lower costs related to manual labeling, rework, scrap, and quality incidents, while enabling higher process transparency and more efficient quality assurance. The ink system and associated marking process can be commercialized as: • A new high-performance ink product line for extreme industrial environments. • A complete marking solution (ink + application guidelines, possibly integration with vision/reading systems) for metal processing industries (e.g. automotive, aerospace, heavy industry). Within 1–2 years after project completion, the partners expect initial sales in domestic markets and subsequent expansion into international niche markets requiring high-temperature traceability. <u>Impact for Korean SMEs and industrial ecosystem</u> The project will equip Korean SMEs with a key enabling technology for digital and data-driven manufacturing. By making each component traceable across all process steps, SMEs can implement robust track-and-trace systems and advanced quality analytics, which are essential elements of smart manufacturing and Industry 4.0. This strengthens the competitiveness of Korean suppliers in global value chains and supports the development of a local ecosystem for high-value, traceability-enabled metal components. <u>Scientific and technological impact</u> Scientifically, the project will advance understanding of high-temperature-stable pigments and binders, ink–metal surface interactions under extreme conditions, and the design of codes/marks that remain machine-readable after thermal and chemical stress. The results will form a basis for future R&D on functional inks and coatings for harsh environments and will be disseminated through joint publications, conference presentations, and, where appropriate, patent applications.

	<u>Environmental and societal impact</u> By reducing scrap rates, rework, and product recalls through improved traceability and process control, the technology contributes to more resource- and energy-efficient production. In addition, reliable automated marking and identification can reduce manual handling and checking of hot or heavy parts, thereby improving occupational safety and lowering the physical burden on workers in Korean SMEs. Potential new use cases include foundries and forging companies (steel and aluminum cast and forged parts), heat treatment shops and furnace manufacturers, pipe and profile manufacturers for pipeline and structural steel, manufacturers of components for energy plants and rail vehicles, and automotive and aerospace suppliers for high-temperature-exposed metal parts. Overall, the joint R&D project will provide Korean SMEs with a robust key technology for high-temperature part traceability and smart manufacturing, while enabling the research institute to transfer and further develop its expertise in high-performance ceramic inks and industrial printing solutions in collaboration with strong industrial partners.

6. References

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(3) Bischoff, P.; Zeh, C.; Schuster, C.; Härtling, T.; Kroh, C. D5.1 Image-Based Predictive Maintenance Concept for Inkjet Printing of Ceramic Inks. Proceedings of SMSI **2021** Measurement Science 2021, 262–263. DOI: 10.5162/SMSI2021/D5.1.

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